

Elements of Statistics and Probability

Nabil Ahmed Uthso

School of Data and Computational Sciences

BRAC University

E-mail: nabil.ahmed@bracu.ac.bd

Table of Contents

1 Regression Analysis

2 Multiple Linear Regression

Introduction

- Regression analysis is the study of dependence and it is used to answer the questions, such as
 - Does changing class size affect success of students?
 - Do changes in diet result in changes in cholesterol level, and if so, do the results depend on other characteristics such as age, sex, and amount of exercise?
 - Do countries with higher per person income have lower birth rates than countries with lower income?

Introduction

- In general, the goal of regression analysis is to summarize observed data as simply, usefully, and elegantly as possible
- Depending on the situations, different regression models are used in statistical analysis and in this course, we will learn **linear regression** models
- In regression analysis, the dependence between a response (e.g. cholesterol level) and one or more predictors (e.g. diet type, age, sex, etc.) is studied

Introduction

- Generally, the notations Y is used for the response and X is for a predictor (X_1, X_2 , so on if there are more than one predictor)
- In regression analysis with one predictor, the data consist of n pairs

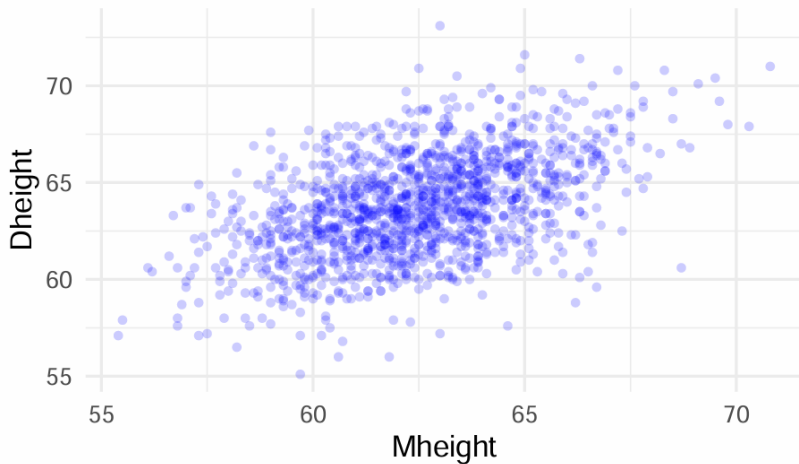
$$\{(y_i, x_i), i = 1, 2, \dots, n\}$$

- y_i and x_i are the values of the response and the predictor obtained from the i^{th} observational unit

Example: Inheritance of Height

- One of the first uses of regression was to study inheritance of traits from generation to generation
- Karl Pearson collected 1375 heights of mothers in the UK (age < 65) and one of their adult daughters (age > 18)
- Interest was in inheritance from the mother to the daughter
 - Mother height $\rightarrow$ predictor
 - Daughter height $\rightarrow$ response
- Questions of interest:
 - Do taller mothers tend to have taller daughters?
 - Do shorter mothers tend to have shorter daughters?

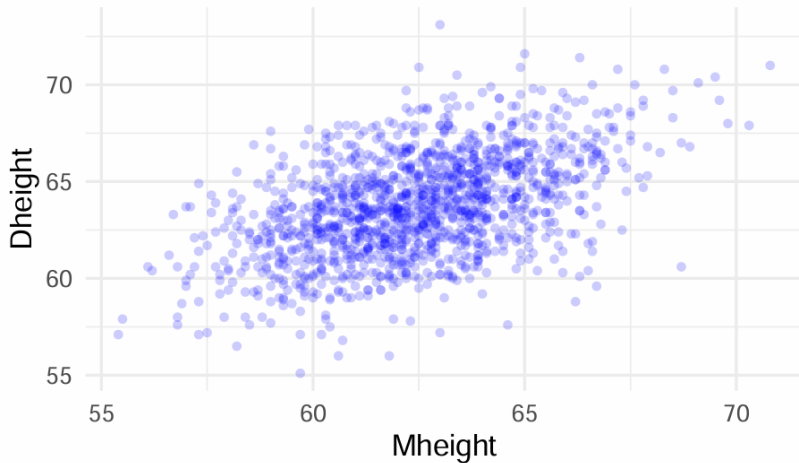
Scatterplot of Mothers and Daughters Height



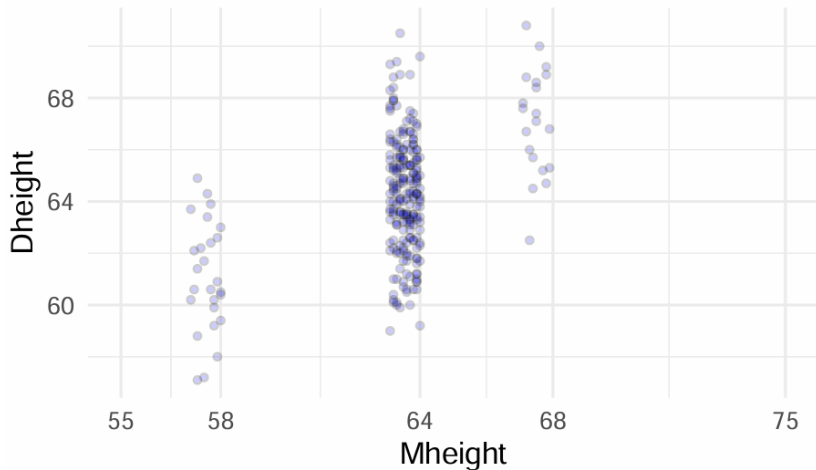
Summary of the Example

- The scatterplot refers to a situation where two quantities are associated with each experimental/observational unit
- In the scatterplot, there appears to be a rough trend or association between the variables
- Although an increasing trend was evident, there was not a one-to-one relationship between the two quantities. It was not true that
 - every taller mother had a taller daughter in comparison with a shorter mother

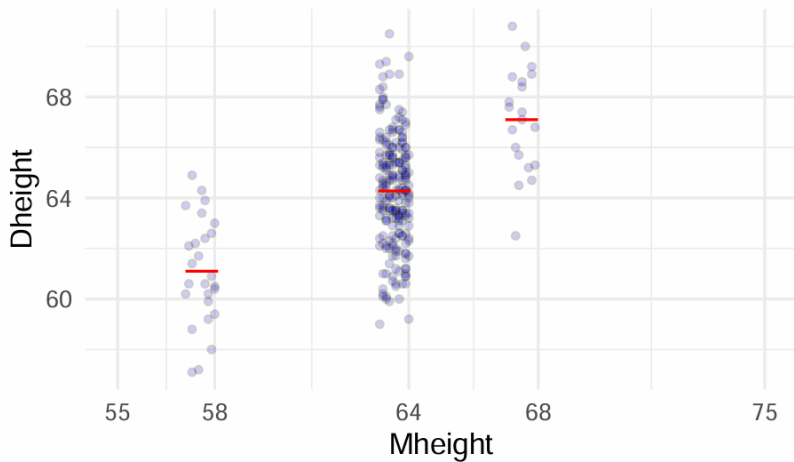
Mean Function



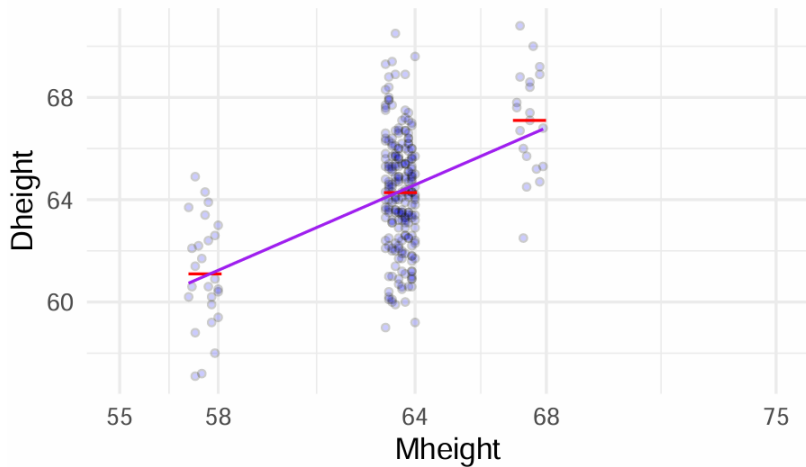
Mean Function



Mean Function



Mean Function



Mean Function

- The mean function of the response Y when the predictor is fixed at a specific value x (say) is defined as

$$E(Y|x) = E(Y|X = x) \rightarrow \text{a function that depends on } x$$

- $E(Y|X = x) \rightarrow$ expected value of Y given $X = x$

Intercept and Slope

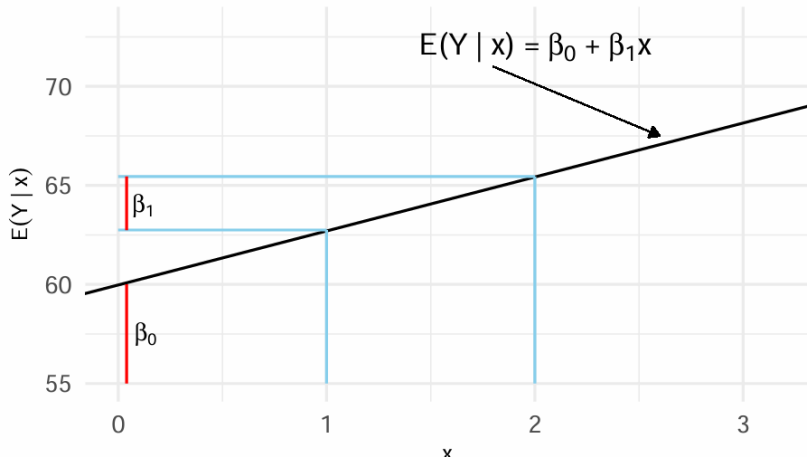
- Different functions of x can be considered for the mean function and the simplest function could be a straight line

$$E(Y|X = x) = \beta_0 + \beta_1 x$$

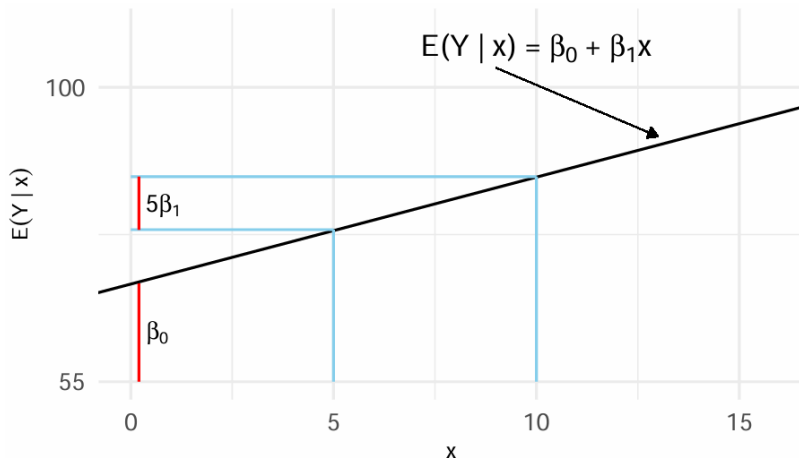
- Mean function depends on two parameters β_0 (intercept) and β_1 (slope), which need to be estimated from the data
- If we assume that the mother's height has no effect on daughter's height, we could consider the following model for the mean function

$$E(Y|X = x) = \beta_0$$

Intercept and Slope



Intercept and Slope



Interpretations of Intercept and Slope

- Mean function: $E(Y|X = x) = \beta_0 + \beta_1 x$
- Interpretation of the intercept β_0 :

$$E(Y|X = 0) = \beta_0$$

- $\beta_0 \rightarrow$ expected **response** when **predictor** is fixed at 0
- Interpretation of the slope β_1 :

$$\begin{aligned} E(Y|X = x + 1) - E(Y|X = x) &= \{\beta_0 + \beta_1(x + 1)\} - \{\beta_0 + \beta_1 x\} \\ &= \beta_1 \end{aligned}$$

- $\beta_1 \rightarrow$ change of the expected responses for one unit increase of the predictor

Regression Model

- In general, the following model can be assumed for the response corresponding to the i^{th} unit

$$\begin{aligned}y_i &= E(Y|x_i) + \epsilon_i \\ &= \{\text{systematic part}\} + \{\text{random part}\}\end{aligned}$$

- $\epsilon_i \rightarrow$ random error term corresponding to the i^{th} response
- **Assumptions** related to the error term:
 - random error, ϵ 's are independent
 - $\epsilon_i \sim N(0, \sigma^2)$

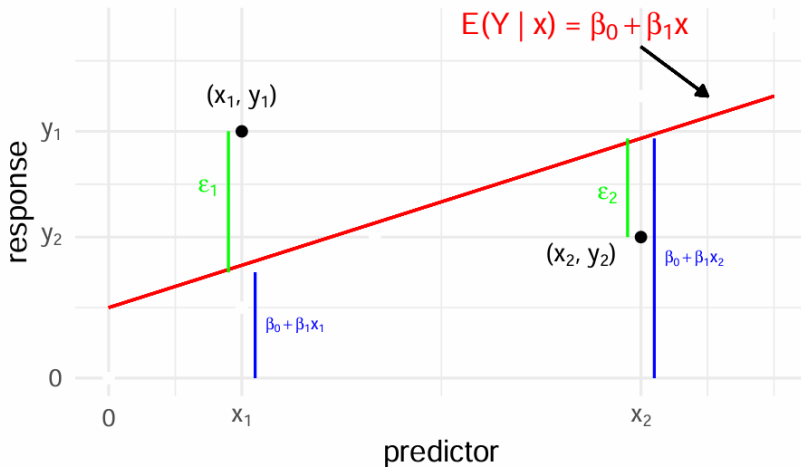
Regression Model

- Using the model of mean function, the model for the i th response becomes

$$\begin{aligned}y_i &= E(Y|x_i) + \epsilon_i \\ &= \beta_0 + \beta_1 x_i + \epsilon_i\end{aligned}$$

- A linear regression model of this form (with one predictor) is known as **simple linear regression** model

Mean Function of Simple Linear Regression Model



Fitted Model

- Regression model for the i^{th} observation:

$$\begin{aligned}y_i &= E(Y|x_i) + \epsilon_i \\&= \beta_0 + \beta_1 x_i + \epsilon_i\end{aligned}$$

- Fitted (estimated) model for the i^{th} observation:

$$\hat{E}(Y|x_i) = \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

- $\hat{y}_i \rightarrow$ estimate of the mean function $E(Y|x_i)$

Residuals

- Random errors ϵ of the model ($y = \beta_0 + \beta_1 x + \epsilon$) are unknown and can be estimated using fitted model
- Estimated errors are known as residuals, which is defined as the difference between response and the corresponding fitted value
 - Estimated error (residual) for the i^{th} observation

$$\hat{\epsilon}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Error Sum of Squares (ESS)

- Simple linear regression model can be expressed in terms of errors:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \implies \epsilon_i = y_i - \beta_0 - \beta_1 x_i$$

- Ordinary least square (OLS) method of estimation is based on the error sum of squares (ESS)

$$ESS = \sum_{i=1}^n \epsilon_i^2 = \sum_{i=1}^n (y_i - \beta_0 - \beta_1 x_i)^2$$

- In OLS method of estimation, ESS is considered as a function of regression parameters β_0 and β_1
- OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ correspond to the minimum of the ESS

OLS Estimators

- OLS estimators:

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \\ &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} \\ \hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x}\end{aligned}$$

- OLS method requires responses are to be independent
- For a given data, $\hat{\beta}_0$ and $\hat{\beta}_1$ corresponds to the minimum of ESS

Residual Sum of Squares (RSS)

- Using the OLS estimators $\hat{\beta}_0$ and $\hat{\beta}_1$

$$\hat{E}(Y|x_i) = \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i \rightarrow \text{fitted model}$$

- Residuals** (estimated errors):

$$\hat{\epsilon}_i = y_i - \hat{y}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$$

Inheritance of Height

- Model of interest “daughter height (y) on mother height (x)”:

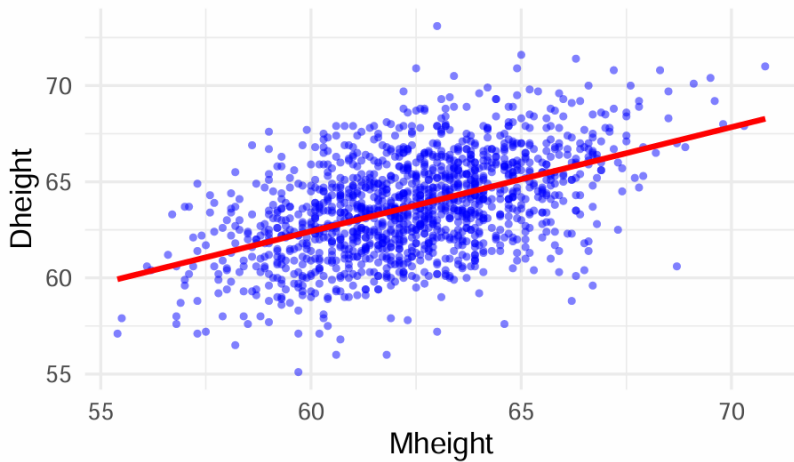
$$y_i = \beta_0 + \beta_1 x_i$$

- Fitted model:

$$\hat{y} = 29.92 + 0.54x$$

- Interpretation of $\hat{\beta}_0 = 29.92$:
- Interpretation of $\hat{\beta}_1 = 0.54$:

Inheritance of Height



Measures of Variation

- **Total sum of squares (TSS):** A measure of variation on the values of the dependent variable (y) around their mean value ($\bar{y}$):

$$TSS = \sum_{i=1}^n (y_i - \bar{y})^2$$

- **Regression sum of squares (RegSS):** Sum of the squared differences between the predicted value of y and the mean value of y :

$$RegSS = \sum_{i=1}^n (\hat{y}_i - \bar{y})^2$$

- **Residual sum of squares (RSS):** Sum of the squared differences between the observed value of y and the predicted value of y :

$$RSS = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Measures of Variation

- Relationship: $TSS = RegSS + RSS$
- Coefficient of determination:

$$R^2 = \frac{\text{Explained variation}}{\text{Total variation}} = \frac{RegSS}{TSS} = \frac{TSS - RSS}{TSS} = 1 - \frac{RSS}{TSS}$$

- Interpretation:**

- R^2 determines the proportion of variation in Y which is explained by independent variable (X) of the regression model
- For example, $R^2 = 0.74$ implies that 74% of the total variation of Y can be explained by the regression model

Example

- A small experiment is conducted involving 10 students to investigate the association between students' math scores and overall scores in exam.

Math score (%):	70	85	66	15	69	49	73	44	88	69
------------------------	----	----	----	----	----	----	----	----	----	----

Overall score (%):	73	90	50	31	56	55	80	58	85	73
---------------------------	----	----	----	----	----	----	----	----	----	----

- (a) Fit a regression model for the association between math score and overall score. Interpret the estimated regression coefficients.
- (b) What is the predicted overall score for a student with a math score 52? Show that $\sum_{i=1}^n \hat{\epsilon}_i = 0$.
- (c) Find the coefficient of determination and interpret the result.

Example (cont.)

x	y	xy	x^2	y^2
70	73	5110	4900	5329
85	90	7650	7225	8100
66	50	3300	4356	2500
15	31	465	225	961
69	56	3864	4761	3136
49	55	2695	2401	3025
73	80	5840	5329	6400
44	58	2552	1936	3364
88	85	7480	7744	7225
69	73	5037	4761	5329

$$\Sigma x_i = 628 \quad \Sigma y_i = 651 \quad \Sigma x_i y_i = 43993 \quad \Sigma x_i^2 = 43638 \quad \Sigma y_i^2 = 45369$$

- $\bar{x} = \Sigma x_i / n = 628 / 10 = 62.8, \quad \bar{y} = \Sigma y_i / n = 651 / 10 = 65.1$

Example (cont.)

- (a) • Estimated slope ($\hat{\beta}_1$):

$$\begin{aligned}\hat{\beta}_1 &= \frac{\sum_{i=1}^n x_i y_i - n\bar{x}\bar{y}}{\sum_{i=1}^n x_i^2 - n\bar{x}^2} \\ &= \frac{43993 - (10 \times 62.8 \times 65.1)}{43638 - (10 \times 62.8^2)} \\ &= 0.74\end{aligned}$$

- **Interpretation of $\hat{\beta}_1 = 0.74$:** For every 1% increase in math score, the overall score increases by an average of 0.74%, **or**, on average, for every 1% increase in math score, the overall score increases by 0.74%.

Example (cont.)

- **Estimated intercept ($\hat{\beta}_0$):**

$$\begin{aligned}\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1 \bar{x} \\ &= 65.1 - (0.74 \times 62.8) \\ &= 18.63\end{aligned}$$

- **Interpretation of $\hat{\beta}_0 = 18.63$:** The intercept of 18.63 represents the expected overall score when the math score equals zero.
- The intercept (18.63) indicates the predicted overall score when math score is 0%. However, since a math score of 0% may fall outside the range of observed data, this value mainly serves as a mathematical constant needed to fit the regression line.

Example (cont.)

- Therefore, the fitted regression model is given by

$$\hat{y} = 18.63 + 0.74x$$

- (b) The predicted overall score for a student with a math score 52:

$$\begin{aligned}\hat{y} &= 18.63 + 0.74x \\ &= 18.63 + (0.74 \times 52) \\ &= 57.11\end{aligned}$$

Example (cont.)

x_i	y_i	$\hat{y}_i$	$\hat{\epsilon}_i$	$\hat{\epsilon}_i^2$
70	73	70.43	2.57	6.60
85	90	81.53	8.47	71.74
66	50	67.47	-17.47	305.2
15	31	29.73	1.27	1.61
69	56	69.69	-13.69	187.41
49	55	54.89	0.11	0.01
73	80	72.65	7.35	54.02
44	58	51.19	6.81	46.38
88	85	83.75	1.25	1.56
69	73	69.69	3.31	10.96

$$\Sigma x_i = 628 \quad \Sigma y_i = 651 \quad - \quad \Sigma \hat{\epsilon}_i = -0.02 \quad \Sigma \hat{\epsilon}_i^2 = 685.5$$

Example (cont.)

- Therefore, $\sum \hat{\epsilon}_i = -0.02 \approx 0$

(c) Coefficient of determination (R^2):

$$\begin{aligned} R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{\sum_{i=1}^n \hat{\epsilon}_i^2}{\sum_{i=1}^n y_i^2 - n\bar{y}^2} \\ &= 1 - \frac{685.5}{45369 - (10 \times 65.1^2)} \\ &= 1 - 0.229 \\ &= 0.771 \end{aligned}$$

- **Interpretation of R^2 :** Approximately 77.1% of the variation in overall scores among these students can be explained by their math scores, according to this linear regression model.

Some Important Relations

- $r = \hat{\beta}_1 \times \sqrt{\frac{S_x^2}{S_y^2}} = \hat{\beta}_1 \times \frac{\sqrt{S_x^2}}{\sqrt{S_y^2}} = \hat{\beta}_1 \times \frac{S_x}{S_y}$
- $r = \hat{\beta}_1 \times \sqrt{\frac{S_{xx}}{S_{yy}}} = \hat{\beta}_1 \times \frac{\sqrt{S_{xx}}}{\sqrt{S_{yy}}} = \hat{\beta}_1 \times \frac{SD_x}{SD_y}$

Exercise 1

- A study was conducted to explore how the number of years of formal education (X) affects monthly income (in thousand BDT) (Y) among young professionals in Dhaka. The fitted model is given by:

$$\hat{y} = 15 + 2x$$

- Additional information: $SD_x = 2.5$ and $SD_y = 10$
- (a) Interpret the estimated regression parameters. Predict the income of a person with 15 years of education.
- (b) Calculate the correlation coefficient and the coefficient of determination. Interpret your findings.

Table of Contents

1 Regression Analysis

2 Multiple Linear Regression

Multiple Linear Regression (MLR)

- Multiple linear regression is a statistical method used to predict the value of a dependent variable based on two or more independent variables
- The equation for multiple linear regression is:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \epsilon$$

- Y is the dependent variable
- $X_1, X_2, \dots, X_p$ are the independent variable
- β_0 is the intercept
- $\beta_1, \beta_2, \dots, \beta_p$ are the coefficients
- ϵ is the error term

Estimated MLR Equation

- The estimated multiple regression equation is as follows

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \dots + \hat{\beta}_p X_p$$

- where $\hat{\beta}_0, \hat{\beta}_1, \dots, \hat{\beta}_p$ are the estimates of the regression parameters, and $\hat{Y}$ is the estimated outcome

R^2 and Adjusted R^2

- Coefficient of multiple determination, R^2 is a number that determines the proportion of total variation in Y which is explained by independent variables of the regression model
- In general, R^2 increases when an independent variable is added to the model, regardless of the contribution of that variable
- Therefore, it is difficult to judge whether an increase in R^2 is really telling us anything important
- So, it is preferred to use adjusted R^2 (R^2_{adj})

Adjusted R^2

- Adjusted R^2 :

$$R_{adj}^2 = 1 - \left\{ \frac{n-1}{n-p} \times (1 - R^2) \right\}$$

- where n is the sample size, p is the number of independent variables, and R^2 is the coefficient of multiple determination
- R_{adj}^2 increases only for potential independent variable but R^2 increases for all variables. So, sometimes R^2 may give us misleading conclusion.

Example

- Suppose the total travel time (Y) (in hours) of a trucking company depends on the number of miles traveled (X_1) and the number of deliveries (X_2). Using the data of ten driving assignments, the estimated multiple regression equation was

found as

$$\hat{Y} = -0.87 + 0.06X_1 + 0.92X_2$$

- (a) Interpret the estimated regression parameters.
- (b) Suppose the $R^2 = 0.90$. Compute the R_{adj}^2 and comment on the goodness of fit of the model.
- (c) Predict the travel time for 85 miles traveled and 5 deliveries.

Example (cont.)

(a) Interpretation:

- $\hat{\beta}_0 = -0.87$: The estimated travel time is -0.87 hours when both the number of miles traveled and the number of deliveries are zero (!).
- $\hat{\beta}_1 = 0.06$: On average, for each additional mile traveled, the travel time increases by 0.06 hours, holding the number of deliveries constant.
- $\hat{\beta}_2 = 0.92$: On average, for each additional delivery, the travel time increases by 0.92 hours, holding the number of miles traveled constant.

Thank You